

5.1.1. Stacked Plot

06:59

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```

1 import matplotlib.pyplot as plt
2 import pandas as pd
    
```

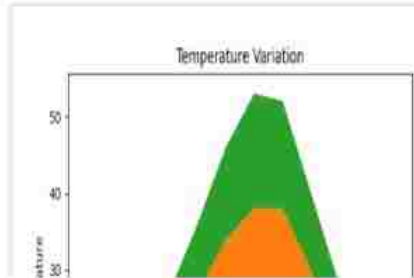
Average time: 0.380 s 380.00 ms

Maximum time: 0.380 s 380.00 ms

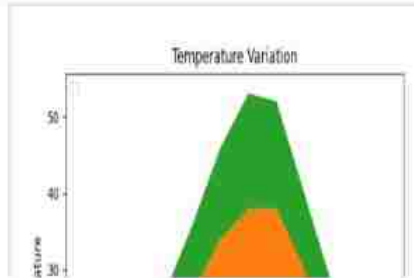
1 out of 1 shown test case(s) passed

Test case 1 380 ms Debug

Expected output



Actual output



Terminal Test cases

5.2.1. Titanic Dataset

02:21

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

titanicDat...

Submit

1 `import pandas as pd`

2 `import matplotlib.pyplot as plt`

Average time

1.659 s

1659.00 ms

Maximum time

1.659 s

1659.00 ms

1 out of 1 shown test case(s) passed

Test case 1

1659 ms

Debug

Expected output

Actual output

Passenger Class Distribution

Gender Distribution

Age Distribution

Survived Status

Passenger Class Distribution

Gender Distribution

Age Distribution

Survived Status

Terminal

Test cases

5.2.2. Histogram of passenger information of Titanic

02:49

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

Histogra...

Submit

1import pandas as pd

2import matplotlib.pyplot as plt

Average time0.459 s459.00 ms

Maximum time0.459 s459.00 ms

1 out of 1 shown test case(s) passed

Test case 1459 ms

Debug

Expected output

Actual output

Age Distribution

Age Distribution

Terminal

Test cases

5.2.3. Bar plot of survival rate of passengers

00:30 [A](#) [L](#) [C](#) [S](#) [P](#)

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d
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BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
```

Average time

0.463 s
463.00 ms

Maximum time

0.463 s
463.00 ms

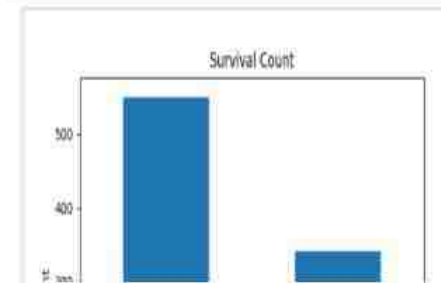
✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 463 ms

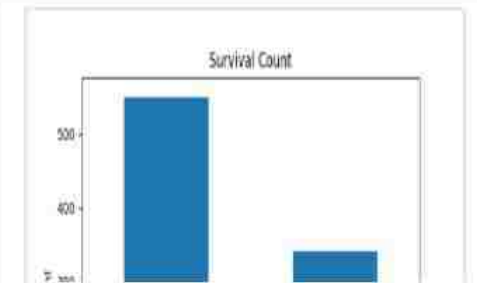
[Debug](#)

[Test cases](#)

Expected output



Actual output



[Terminal](#)

[Test cases](#)

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- 1. Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- 2. Use a **stacked bar chart** to display the survival counts.
- 3. Add the title "**Survival by Gender**" to the chart.
- 4. Label the x-axis as '**Gender**' and the y-axis as '**Count**'.
- 5. The legend should indicate '**Not Survived**' and '**Survived**'.

The Titanic dataset contains columns as shown below,

P as se ng er1	S ur vi ve	Pc la ss	N a m e	S ex	A ge	Si b S n	P ar ch	Ti ck et	Fa re	C ab in	E m ba rk
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BarPlotOf... Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt

Average time: 0.594 s (594.00 ms) | Maximum time: 0.594 s (594.00 ms) | 1 out of 1 shown test case(s) passed

Test case 1 (594 ms) Debug

Expected output:

Actual output:

Terminal | Test cases

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

- 1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
- 2. Use a **stacked bar chart** to display the survival counts.
- 3. Add the title "**Survival by Pclass**" to the chart.
- 4. Label the x-axis as '**Pclass**' and the y-axis as '**Count**'.
- 5. The legend should indicate '**Not Survived**' and '**Survived**'.

The Titanic dataset contains columns as shown below,

P	S	Pc	N	S	A	Si	P	Ti	Fa	C	E
a	u	l	a	e	g	b	a	c	r	a	m
s	r	a	m		e	S	r	k	e	b	a
s	v	s	e			p	ch	et		in	r
e	i										k
d	d										e

BarPlotOf...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

Average time

0.515 s

515.00 ms

Maximum time

0.515 s

515.00 ms

1 out of 1 shown test case(s) passed

Test case 1

515 ms

Debug

Expected output

Actual output

Survival by Pclass

Survival by Pclass

Terminal

Test cases

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked**" to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

P	S		N			Si	P	Ti		C	E
as	ur	Pc									m

BarPlotOf... Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt

Average time 0.605 s 605.00 ms
Maximum time 0.605 s 605.00 ms
1 out of 1 shown test case(s) passed

Test case 1 605 ms Debug

Expected output

Actual output

Terminal Test cases

5.2.7. Box plot for Age Distribution 00:36

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- 1. Use the **Pclass** column to group the data for the boxplot.
- 2. Set the title of the plot to **"Age by Pclass"**.
- 3. Remove the default subtitle with **plt.suptitle("")**.
- 4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

BoxPlotF...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

Average time 0.545 s 545.00 ms

Maximum time 0.545 s 545.00 ms

1 out of 1 shown test case(s) passed

Test case 1 545 ms Debug

Expected output

Actual output

Age by Pclass

Age by Pclass

Terminal Test cases

5.2.8. Box Plot for Age by Survived

00:43

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d
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BoxPlotF...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

Average time0.584 s584.00 ms

Maximum time0.584 s584.00 ms

1 out of 1 shown test case(s) passed

Test case 1584 ms

Debug

Expected output

Actual output

Age by Survival

Age by Survival

Terminal

Test cases

5.2.9. Box Plot for Fare by Pclass

00:32

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

BoxPlotF...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

Average time0.519 s519.00 ms

Maximum time0.519 s519.00 ms

1 out of 1 shown test case(s) passed

Test case 1519 ms

Debug

Expected output

Actual output

Terminal

Test cases

5.2.10. Scatter Plot for Age vs. Fare 00:34

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

- 1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- 2. Set the title of the plot to **"Age vs. Fare"**.
- 3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d

AgeFareS... Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt

Average time 0.409 s 409.00 ms
Maximum time 0.409 s 409.00 ms
1 out of 1 shown test case(s) passed

Test case 1 409 ms Debug

Expected output

Actual output

Terminal Test cases

5.2.11. Scatter Plot for Age vs. Fare by Survived

00:47

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

P a s s e n g e r I d	S u r v i v e d	P c l a s s	N a m e	S e x	A g e	S i b S p	P a r c h	T i c k e t	F a r e	C a b i n	E m b a r k e d
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AgeFareS...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

Average time

0.447 s

447.00 ms

Maximum time

0.447 s

447.00 ms

1 out of 1 shown test case(s) passed

Test case 1 447 ms

Debug

Expected output

Actual output

Age vs. Fare by Survival

Age vs. Fare by Survival

Terminal

Test cases